S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 10

Aim: Apply prompt engineering to construct a prompt for learning the concept of Adversarial Search.

Name of Student:

Roll No.:

Semester/Year: V/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: To design and refine an effective prompt using prompt engineering techniques for learning the concept of Adversarial Search from Artificial Intelligence.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To be able to understand the concept of a *prompt* (input given to an AI model).
- To be able to understand the concept of *prompt engineering* (designing inputs to get desired outputs)
- To be able to work with basic idea and models like ChatGPT.
- To be able to distinguish between a vague prompt and a well-structured one.

THEORY:

i. Fundamentals of Prompt Engineering

Prompt engineering is the process of designing and structuring inputs (prompts) to get accurate, relevant, and well-formatted outputs from AI systems such as ChatGPT. Prompt engineering is how you ask a question to AI so that it gives the best possible answer.

ii. Characteristics of a Good Prompt

A good prompt is based on the following core principles:

- **Clarity** → Avoid ambiguity
- **Specificity** → Define exact task
- **Context** → Provide background
- **Constraints** → Limit output (length, format, style)
- **Intent** → Clearly state what is required

Example:

- Weak: *"Explain AI"*
- Strong: *"Explain AI in 5 bullet points with real-world examples for beginners"*

iii. Prompt Patterns / Techniques

The common patterns of the prompt along with their usage are as follows:

a) Role-Based Prompting

"You are an AI professor…"

b) Step-by-Step Prompting

"Explain step-by-step…"

c) Few-shot Prompting

Provide examples in the prompt

d) Instruction + Constraints

"Explain in 100 words using simple language"

e) Chain-of-Thought (Guided Reasoning)

"Show reasoning before final answer"

iv. Task-Oriented Prompting

Based on the tasks we can have different ways to provide the prompt. Following are the examples to design prompts for different tasks:

- **Learning/Explanation**
- **Coding**
- **Summarization**
- **Question generation**
- **Problem solving (AI, Cloud, etc.)**

v. Iterative Refinement (Very Important)

The first prompt may not be the best one. So, we must learn how the prompt can be refined to get most appropriate outcome:

- First prompt ≠ best prompt
- Improve by:
  - Adding constraints
  - Asking follow-up questions
  - Correcting output

Concept:

Prompt → Output → Refine → Better Output

vi. Evaluation of AI Output

It is very important to evaluate the output generated by AI. Critical thinking should be done to find the answer to the following:

- Is the answer **correct**?
- Is it **complete**?
- Is it **biased or vague**?
- Does it follow instructions?

vii. Ethical and Responsible Use

- Avoid Misuse and Plagiarism
  - Do not copy AI-generated content blindly for assignments or research
  - Always ensure originality and proper attribution when required
- Verify Information
  - AI tools like ChatGPT may generate incorrect or outdated information
  - Cross-check important facts before using them in academic or professional work
- Use AI as a Support Tool
  - AI should assist learning, not replace critical thinking
  - Students should understand concepts rather than depend entirely on AI-generated answers

viii. Basic Applications in Academics

- Generate notes
- Create MCQs and scenario-based questions
- Debug code
- Explain complex topics simply

ix. Perform the following tasks:

- Convert a bad prompt → good prompt
- Generate:
  - Notes
  - Quiz questions
  - Code snippets
- Design a prompt for:
  - Adversarial Search explanation

EXPECTED OUTPUT:

- Definition
- Game tree explanation
- Minimax working
- Alpha-Beta optimization
- Example
- Complexity analysis

INPUT & OUTPUT:

Attempt	Prompt Quality	Output Quality	Improvements
1(Basic)	"Explain Pac-Man in AI."	Too vague, missing adversarial concept	Add adversarial search context
2(Improved)	"Explain Pac-Man as an adversarial search problem with minimax and example."	Better but lacks multi-agent explanation	Add ghosts as multiple MIN agents
3(Refined)	"Explain Pac-Man as an adversarial search problem in AI. Include definition, multi-agent game tree, minimax working, alpha-beta pruning, example with utility values, and complexity analysis in structured format."	High-quality, complete, structured answer	Final optimized prompt

CONCLUSION:

DISCUSSION QUESTIONS:

1. What is prompt engineering?

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2. Why is prompt refinement important?

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3. What is Alpha-Beta pruning?

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4. Difference between weak and strong prompts?

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REFERENCES:

- https://learnpromptengineering.dev
- https://www.linkedin.com/pulse/40-prompt-engineering-resources-upskilling-spencer-taylor-xje0e/
- https://justprompting.in/